Models with Ill-Mannered Error

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Assumptions so far…

$$Data = Model + Error$$

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} \ldots + \beta_n x_{ni} + \varepsilon_i$$

- In these models, we have made important assumptions about the residuals (ε's).
  1. They are independent of each other. ***(independence assumption)***
  2. They are normally distributed. ***(normality assumption)***
  3. They have a constant variance. ***(homoscedasticity/homogeneity of variance assumption)***

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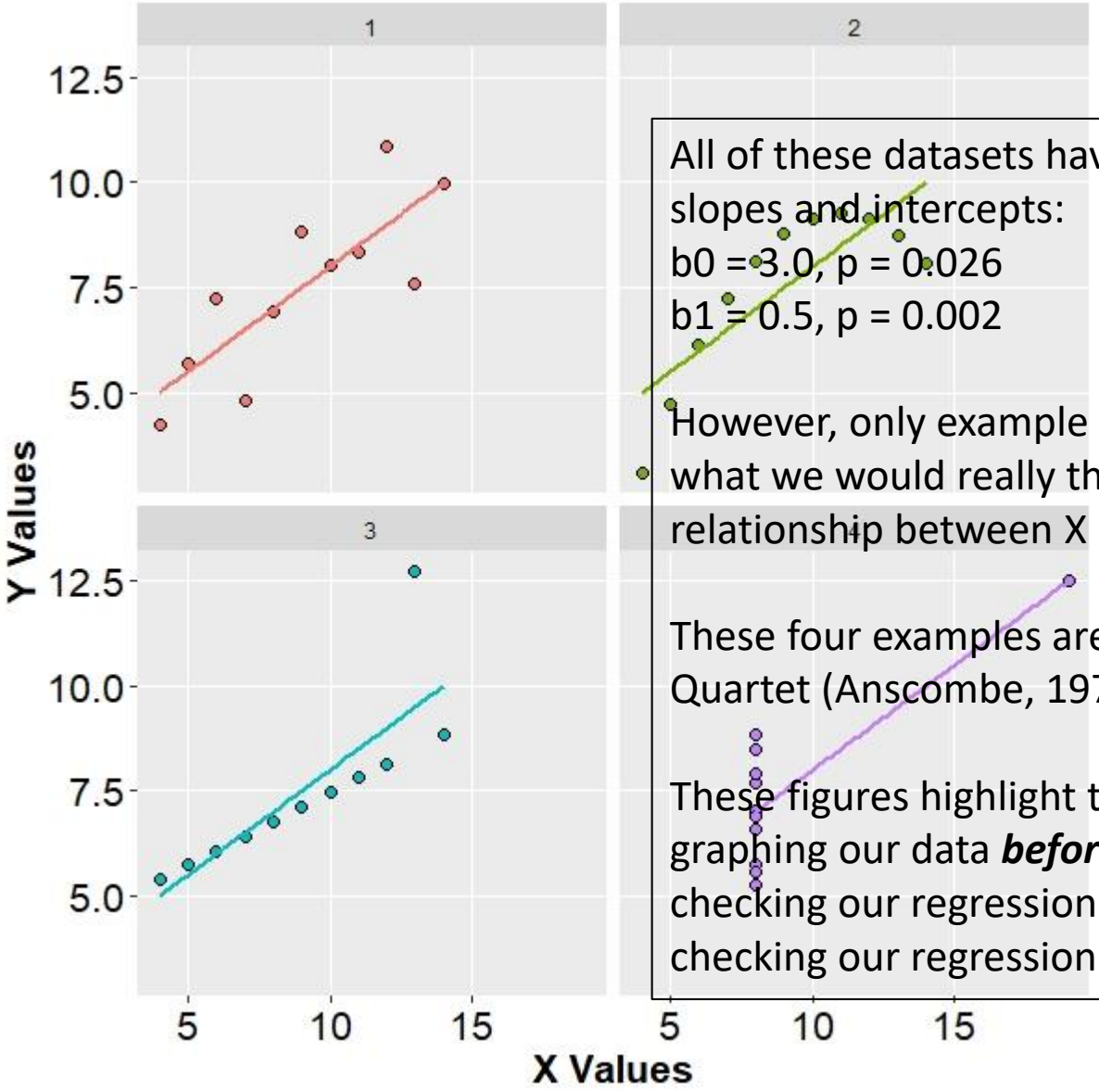

All of these datasets have exactly the same slopes and intercepts:
b0 = 3.0, p = 0.026
b1 = 0.5, p = 0.002

However, only example Number 1 represents what we would really think of as a “fair” linear relationship between X and Y.

These four examples are known as Anscombe’s Quartet (Anscombe, 1973).

These figures highlight the importance of graphing our data ***before*** we analyze it, checking our regression ***diagnostics***, and checking our regression ***assumptions***.

Regression Diagnostics and Statistical Assumptions

- **<u>Diagnostics (checking for "*Idiosyncratic*" Errors):</u>**
  - Leverage (extreme values in X)
  - Outliers (extreme values in Y)
  - Influence (data-points that substantially affect the model)

- **<u>Assumptions (checking for "*Systematic*" Errors):</u>**
  - Normality
  - Homosckedasticity/Homogeneity of Variance